Three factor-base oracle leads

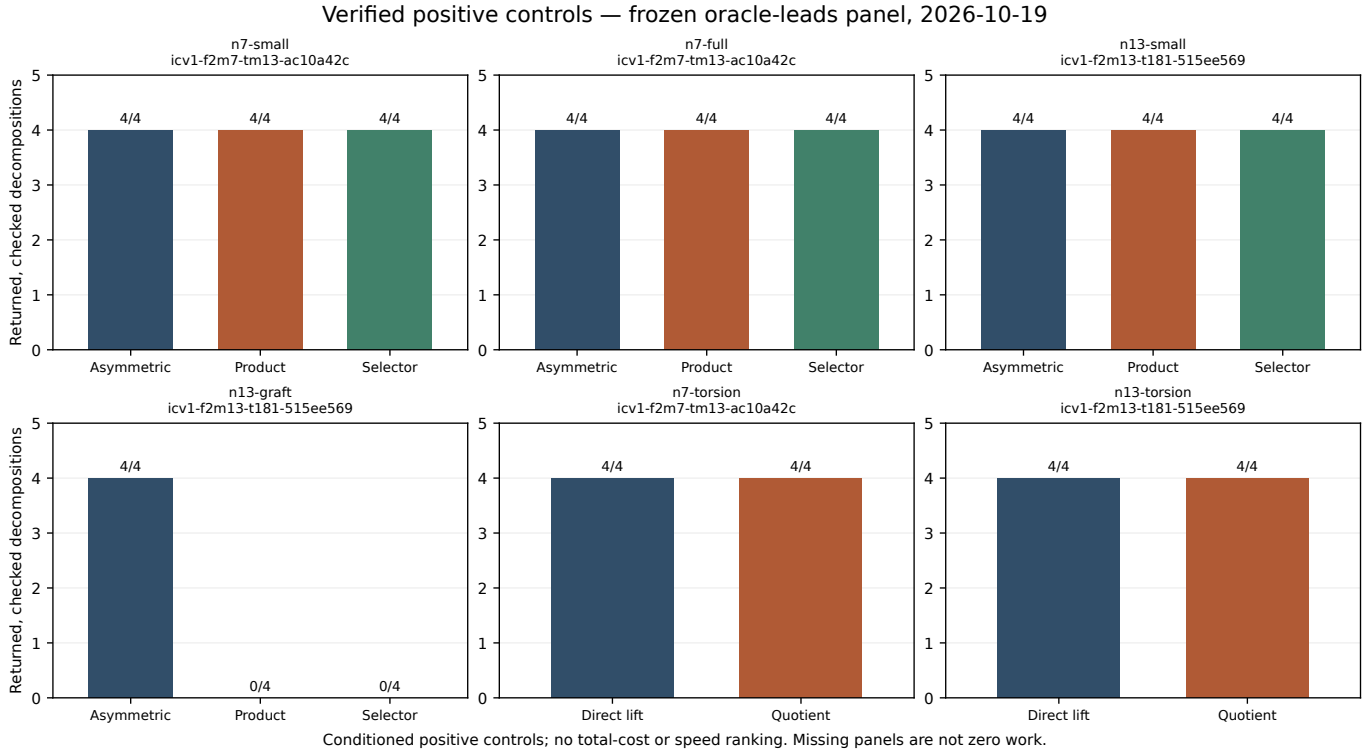
Asymmetric charts, selectors and checked torsion quotients
2026-10-19 · native bounded run and results report

Actual point decompositions, with limits retained

The frozen experiment contains six cases on two exact models. It records 472 bounded trial/arm rows. Separate native field/group arithmetic checks 143 returned original-curve witnesses and 355 algebraic roots. Every emitted witness sums to its original target. Missing completions remain missing; no total-cost winner is admitted.

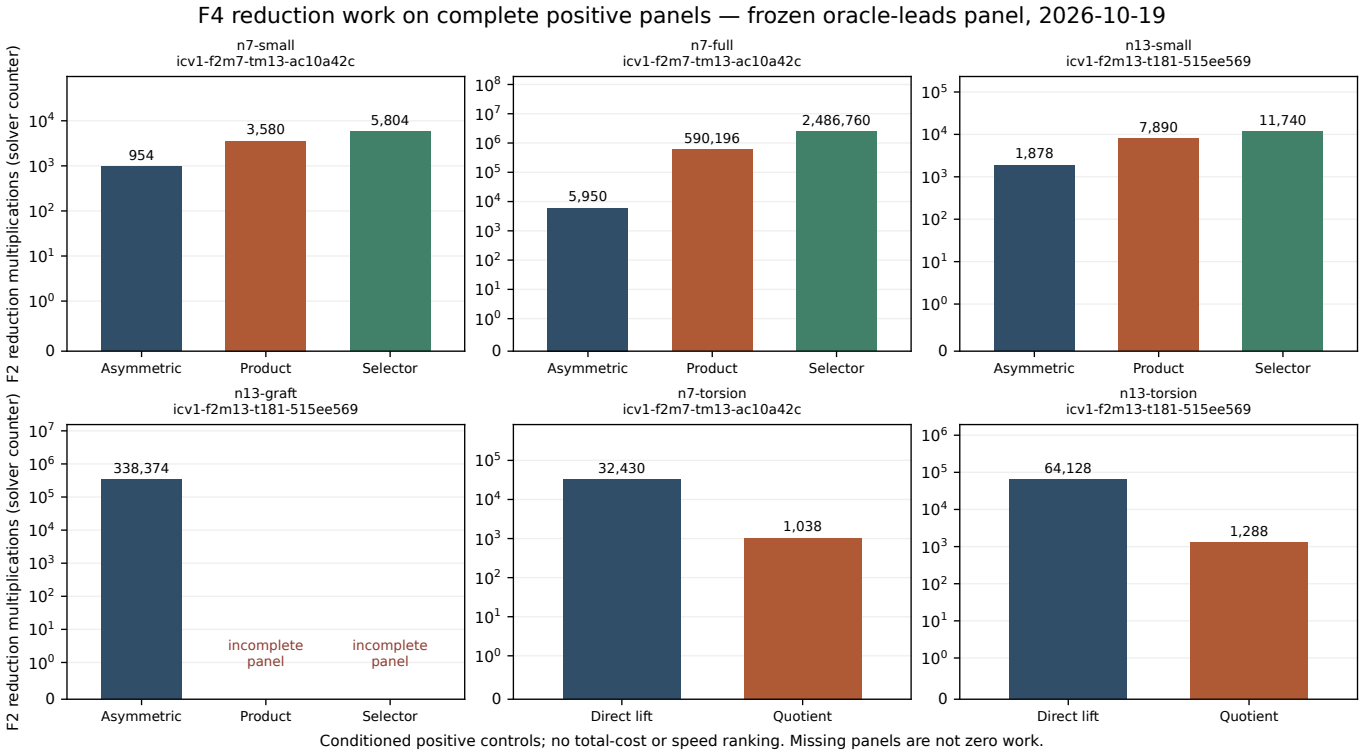
Exact models: `icv1-f2m7-tm13-ac10a42c` (r=71, h=2) and `icv1-f2m13-t181-515ee569` (r=2003, h=4). EC1, full ICV1, field polynomial, generator and source FB1 identities are retained in the copied input files. The two quotient controls use C={±G, ±[2]G}; they are mechanism controls, not selected large-field factor bases or evidence of unknown-scalar recovery.

Case	A points	B points	Union	Projected cols	Support
n7-small	4	2	6	3	10
n7-full	12	8	18	9	50
n13-small	4	2	6	3	10
n13-graft	24	4	28	14	242
n7-torsion	8	8	8	2	8
n13-torsion	8	8	8	2	8



Positive controls are conditioned supported targets, not yield samples. All random arms share 12 training targets and 12 disjoint holdout targets. Controls and holdout cannot increase training rank.

Successful solver-component work



The bars count F2 row-reduction multiplications from the unmodified native F4 engine. A bar is shown only for a completely successful positive panel; an incomplete panel is explicitly omitted, not assigned zero work. Extra selector variables and every attempted asymmetric branch are retained.

This counter excludes parts of symbolic preprocessing, finite-root search, allocation, hashing, setup, verification and rank. F4 solving includes a substitution/enumeration procedure; its productive degree is not certified degree of regularity. The positive panels are tiny and conditioned, and no statistical significance or large-field extrapolation is claimed.

Every call uses grevlex, max degree4, a 750ms cooperative budget, a 2000ms hard allowance per permitted branch, one Rayon thread, one million observed matrix cells and a 1024MiB Linux address-space limit. Hard-stopped counters are incomplete and stay in raw events. No budget or assertion was relaxed.

Positive panels, including the A/A repeat

Case	Arm	Hits	F4 mults	Encoding products	Limits	Degree
n7-small	asymmetric	4/4	954	140	0	2
n7-small	product	4/4	3580	292	0	2
n7-small	reference-repeat	4/4	3580	292	0	2
n7-small	selector	4/4	5804	308	0	2
n7-full	asymmetric	4/4	5950	894	0	2
n7-full	product	4/4	590196	5476	0	3
n7-full	reference-repeat	4/4	590196	5476	0	3
n7-full	selector	4/4	2486760	4148	0	3
n13-small	asymmetric	4/4	1878	140	0	2
n13-small	product	4/4	7890	324	0	2
n13-small	reference-repeat	4/4	7890	324	0	2
n13-small	selector	4/4	11740	324	0	2
n13-graft	asymmetric	4/4	338374	10186	0	3
n13-graft	product	0/4	partial	partial	4	—
n13-graft	reference-repeat	0/4	partial	partial	4	—
n13-graft	selector	0/4	partial	partial	4	—
n7-torsion	direct-lift-base	4/4	32430	944	0	2
n7-torsion	reference-repeat	4/4	32430	944	0	2
n7-torsion	torsion-quotient	4/4	1038	112	0	2
n13-torsion	direct-lift-base	4/4	64128	904	0	2
n13-torsion	reference-repeat	4/4	64128	904	0	2
n13-torsion	torsion-quotient	4/4	1288	112	0	2

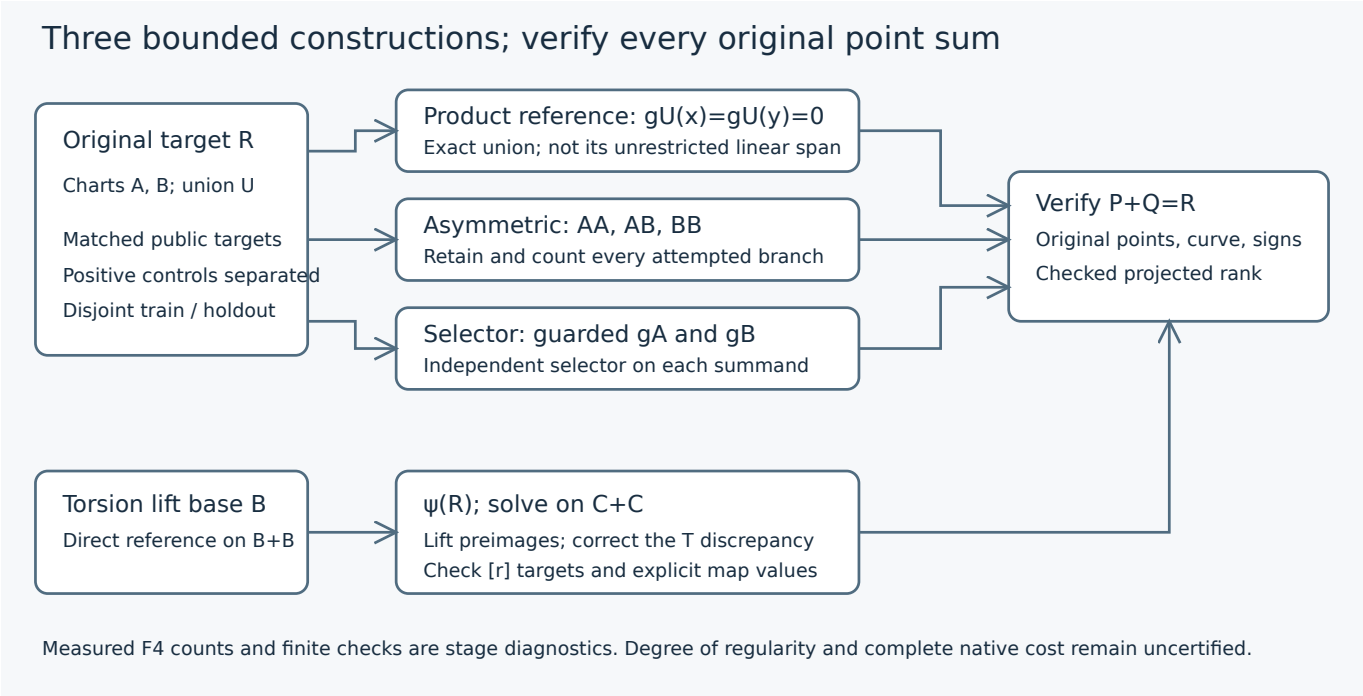
Fresh random probes and projected training rank

All arms receive the same original point targets within each case. Training and holdout are disjoint, but bases were selected using earlier studies. There is no claim of an untouched discovery population. Rank uses distinct nonzero signed cofactor projections: kernel translates do not create fake independent unknowns. No point logarithms are solved.

Case	Arm	Phase	Hits	Absent	Unresolved	Limits	Rank gain
n7-small	asymmetric	holdout	0/12	12	0	0	0
n7-small	product	holdout	0/12	12	0	0	0
n7-small	selector	holdout	0/12	12	0	0	0
n7-small	asymmetric	train	3/12	9	0	0	3
n7-small	product	train	3/12	9	0	0	3
n7-small	selector	train	3/12	9	0	0	3
n7-full	asymmetric	holdout	7/12	5	0	0	0
n7-full	product	holdout	7/12	5	0	0	0
n7-full	selector	holdout	7/12	5	0	0	0
n7-full	asymmetric	train	9/12	3	0	0	6
n7-full	product	train	9/12	3	0	0	6
n7-full	selector	train	9/12	3	0	0	6
n13-small	asymmetric	holdout	0/12	12	0	0	0
n13-small	product	holdout	0/12	12	0	0	0
n13-small	selector	holdout	0/12	12	0	0	0
n13-small	asymmetric	train	0/12	12	0	0	0
n13-small	product	train	0/12	12	0	0	0
n13-small	selector	train	0/12	12	0	0	0
n13-graft	asymmetric	holdout	0/12	12	0	0	0
n13-graft	product	holdout	0/12	1	0	11	0
n13-graft	selector	holdout	0/12	0	0	12	0
n13-graft	asymmetric	train	0/12	12	0	0	0
n13-graft	product	train	0/12	0	0	12	0
n13-graft	selector	train	0/12	0	0	12	0
n7-torsion	direct-lift-base	holdout	2/12	10	0	0	0
n7-torsion	torsion-quotient	holdout	2/12	10	0	0	0
n7-torsion	direct-lift-base	train	2/12	10	0	0	2
n7-torsion	torsion-quotient	train	2/12	10	0	0	2
n13-torsion	direct-lift-base	holdout	1/12	11	0	0	0
n13-torsion	torsion-quotient	holdout	1/12	11	0	0	0
n13-torsion	direct-lift-base	train	0/12	12	0	0	0
n13-torsion	torsion-quotient	train	0/12	12	0	0	0

Absence is accepted only when it agrees with exhaustive original-inventory point truth. An exhausted F4 call is not a refutation. A hard resource stop is not a completed solver call. Full systems, roots, target scalars, witnesses, partial events and independent audit fields remain available.

What the three paths actually do



Asymmetric: solve A+A, A+B, B+B in fixed size-product order, stopping only after a checked witness. Commutativity covers B+A; no same-base branch is silently omitted. Compare against the same union's product-membership arm.

Selector: use an independent Boolean selector for each summand and guard both component membership equations. The accepted domain is exactly the union, not its unrestricted linear span. Both encodings still need actual point witnesses after S3 roots.

Quotient: evaluate dual Frobenius as doubling followed by inverse Frobenius. Solve on C+C at the mapped target, lift through complete preimages, correct the small-torsion discrepancy and verify P+Q=R. The cofactor4 case does not assume membership in 2E implies membership in the chosen prime subgroup. Setup, transport and lifting are not free.

Native CLI

```
factor-base-sweep compare-leads \
  --plan research/factor_base_oracle_leads_20261019/plan.json \
  --out runs/oracle-leads
factor-base-sweep verify --run runs/oracle-leads
```

The executable runs its own native workers. No Cargo launcher is used at runtime. Comparison outputs are write-once; use a new directory, never append to this frozen run. Old executables remain necessary for byte-bound resumes.

Validation, accounting and unresolved claims

Executed source/protocol/input lock: `aaf9e246a3af4be371d7089dc86fc32c61c2f5ac`. The director compiled a hash-bound card; the mechanical tier ran its exact commands; the director assessed the verbatim relay. Canonical check-relay returned exit1 because reloaded output keys sort differently; separate read-only verification confirms exact runner-order bytes and all bindings. The failed canonical result remains retained. Tool elapsed times are resource/receipt observations, not accepted benchmark timings.

One complete serial release-library command: 4400 passed, 0 failed, 118 existing ignores, zero filtered. The unchanged library binary was reused, not freshly compiled. Its tracked dependencies and exact embedded traits fixture are hash-bound in the raw binding receipt. The final CLI has 12 passing unit tests and seven real subprocess integration tests; independent audit has five passing tests. Strict CLI/audit Clippy, all-target compilation and nine CI-scope tests pass.

Independent replay: 2493 sealed members, 691 case-wise unordered pair checks and 16 quotient preimage checks. Separate arithmetic on this same host is not an independent-host certificate or a second F4 engine.

Native-operation weight	Complete total cost	Total-cost ordering
0	Unknown	Unknown
0.1	Unknown	Unknown
1	Unknown	Unknown

This symmetric band applies to references too. Missing emissions and calibration prevent ranking every arm. Magma F4/F5, msolve/FGb and applicable Singular engines were not found on PATH; the matched external comparison is unrun. No fastest implementation, accepted speedup, full-DLP S, rho win, certified regularity or83/130-bit transfer is claimed.

Conductor returned127 and its pinned workflow404. Lean/Lake returned127. These are unavailable checks, not passes. Authoring compilation/formatting failures and all research resource stops are preserved. No historical evidence, draft card, root dependency graph, solver limit or old assertion was rewritten. No S3 or new paid host.

Evidence and prior work

The source-linked README, PROTOCOL.md, PRICING.md, VALIDATION.md, FORMALIZATION.md, copied inputs, card/receipt/assessment and raw archive define the run. Analysis is generated natively from the extracted records; Python renders recorded fields only. Existing scoreboard/leaderboard/browser data are unchanged: there is no admitted IC/rho ratio, newly priced phase, cross-method session, registry addition or complete pipeline result.

Prior quotient and selector derivation: `research/sat_factor_base_review_20260908/continuation-01/EXACT_FACTOR_BASE_RESULTS.md`. Chart input provenance: `research/factor_base_chart_complements_20261019/README.md`. Primary torsion-symmetry context and access limits are documented in LITERATURE.md. All source/evidence hashes and reproduction commands remain linked from this study; existing draft follow-ups stay draft and unrun.